

CivicLens – Citizen-Powered Civic Intelligence Platform

Tagline

"We don't file complaints. We make them impossible to ignore."

1. Vision

CivicLens is a citizen-powered civic intelligence platform that creates a public, verifiable record of infrastructure issues across cities.

Unlike traditional complaint applications that depend on government workflows, CivicLens focuses on transparency, visibility, community validation, and civic intelligence.

The platform allows citizens to report civic problems through WhatsApp or a web portal. AI automatically classifies issues, geotags them, and publishes them on a public civic map.

As reports accumulate, CivicLens generates locality-level Civic Health Scores and identifies emerging infrastructure hotspots.

2. Problem Statement

Current civic complaint systems face several limitations:

- High reporting friction
- Duplicate complaints
- Lack of public visibility
- Low citizen engagement
- Limited accountability
- No locality-wide civic intelligence

Most systems focus on complaint submission rather than creating public awareness and actionable civic insights.

3. Solution

CivicLens creates a public civic intelligence layer.

Citizens submit photos of civic issues.

AI automatically:

- Detects issue type
- Generates structured reports
- Removes duplicate submissions
- Maps issues geographically
- Calculates locality scores

Community members verify reports through upvotes and confirmations.

The platform builds a continuously updated public map of civic conditions.

4. Target Users

Primary Users

- Citizens
- Students
- Community volunteers
- Resident welfare associations
- Local activists

Secondary Users

- NGOs
 - Journalists
 - Researchers
 - Urban planners
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5. Core Features

Feature 1: WhatsApp Reporting Bot

User sends:

- Photo
- Optional text

AI performs:

- Image analysis

- Issue classification
- Description generation
- Geo-location extraction

Example:

User sends pothole image.

System returns:

Detected Issue: Pothole

Severity: High

Location: MG Road

Report Published Successfully

Feature 2: Public Civic Feed

All reports become publicly visible.

Each report contains:

- Image
 - Issue category
 - Location
 - Timestamp
 - Verification count
 - Severity score
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Feature 3: Crowd Verification System

Community validates reports.

Users can:

- Upvote issue
- Confirm issue still exists
- Mark duplicate

Verification score increases report credibility.

Feature 4: Civic Health Score

Every locality receives a dynamic score.

Scoring factors:

- Number of active issues
- Severity distribution
- Verification count
- Issue density

Example:

Area	Score
Ward A	88
Ward B	62
Ward C	37

Feature 5: AI Duplicate Detection

System identifies:

- Same issue
- Nearby reports
- Similar images

Instead of creating duplicates:

- Reports merge
- Verification count increases

Feature 6: Civic Issue Heatmap

Map displays:

- Complaint clusters
- Infrastructure hotspots
- High-risk zones

Users can explore by:

- Locality

- Category
 - Severity
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6. User Flow

WhatsApp Flow

Citizen



Send Photo



AI Classification



Location Extraction



Report Generation



Published to Civic Feed



Community Verification



Civic Score Updated

Web Portal Flow

Citizen



Upload Image



AI Analysis



Preview Report



Publish



Community Verification



Map & Dashboard Update

7. AI Components

Image Classification

Detect:

- Potholes
- Garbage
- Water leakage
- Broken streetlights
- Open manholes
- Road cracks
- Damaged footpaths
- Illegal dumping

Models:

- YOLOv8
 - YOLOv11
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Duplicate Detection

Techniques:

- Image embeddings
 - Geospatial proximity
 - Timestamp matching
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Severity Assessment

Inputs:

- Issue type
- Location context
- Verification count

Outputs:

- Low
 - Medium
 - High
 - Critical
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8. MVP Scope

Included

- WhatsApp Bot
- Website Upload Portal
- AI Classification
- Public Civic Feed
- Crowd Verification
- Civic Health Score
- Heatmap Dashboard

Excluded

- Government integrations
- Resolution tracking
- Multilingual support
- Mobile applications
- Predictive analytics
- Revenue modules

9. Technical Architecture

Option A: WhatsApp First Architecture

Frontend

WhatsApp Business Interface

Messaging Layer

WhatsApp Business Cloud API

Backend

FastAPI

AI Layer

YOLOv8 / YOLOv11

OpenCV

Database

PostgreSQL

Storage

AWS S3

Maps

Leaflet.js

OpenStreetMap

Deployment

AWS EC2

or

Render

Architecture Flow

User



WhatsApp Bot



FastAPI Backend



AI Classification Engine



PostgreSQL



Public Dashboard

10. Low-Cost Alternative (No WhatsApp Cost)

Frontend

React

Next.js

Tailwind CSS

Leaflet.js

Backend

FastAPI

Python

Database

PostgreSQL

Storage

AWS S3

or

Cloudinary

AI Layer

YOLOv8

OpenCV

Hosting

Frontend

Vercel

Backend

Render

Database

Supabase PostgreSQL

11. Website Pages

Home

- Civic Feed
- Upload Issue

- Civic Score Overview
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Map View

- Heatmap
 - Issue Clusters
 - Filters
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Locality Dashboard

- Score
 - Active Issues
 - Top Categories
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Issue Details

- Images
 - Verification Count
 - Timeline
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12. Placeholder Dataset

1. Large pothole on MG Road
 2. Garbage overflow near Bus Stop
 3. Broken streetlight at Market Road
 4. Water leakage near School Junction
 5. Open manhole near Hospital
 6. Damaged footpath near Park
 7. Illegal waste dumping area
 8. Broken traffic signal
 9. Road crack near Railway Crossing
 10. Overflowing drainage channel
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13. Success Metrics

- Reports Submitted
- Verified Reports
- Active Users

- Duplicate Reduction Rate
 - Localities Covered
 - Civic Score Accuracy
 - Map Coverage Density
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14. Future Roadmap

Phase 2

- Predictive civic intelligence
- Time-series trend analysis
- Infrastructure deterioration forecasting

Phase 3

- NGO dashboard
- Journalist portal
- Public APIs

Phase 4

- City-wide digital twin
 - Civic risk forecasting
 - Infrastructure analytics platform
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15. Project Deliverables

- WhatsApp Reporting Bot
 - AI Classification Engine
 - Public Civic Feed
 - Interactive Civic Map
 - Verification System
 - Civic Health Score Dashboard
 - Technical Documentation
 - Deployment Guide
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One-Line Pitch

CivicLens is a citizen-powered civic intelligence platform that uses AI and community verification to create a public, real-time map of urban infrastructure issues and civic health.